Representational Measurement of Similarity: The Additive Difference Model, Revisited

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In dieser Ausarbeitung geht es darum, wie man eine Metrik auf ordinalen Ähnlichkeitsdaten herleiten kann, die zusätzlich additiv ist und durch das additive difference model repräsentiert werden kann. Das Hauptresultat ist, dass die l^p Metrik die einzige additive-difference Metrik, die additiv auf Liniensegmenten ist, was eng mit der Homogenität dieser Metrik zusammenhängt. Zum Herleiten dieses Hauptresultates werden die Grundelemente von Messtheorie, sowie von proximity measurement mithilfe von geometrische Modellen eingeführt. Diese Ausarbeitung basiert auf dem Paper Tversky and Krantz, 1970 und den Büchern Suppes, 2014 und Breinbauer, 1976.

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5.1.1 Convexity of δ Balls

Theorem 5.19 (Eindeutigkeitssatz zu l^p Metriken).
Suppose $\langle A, \succsim \rangle$ is a complete, segmentally additive, factorial proximity structure, that is also and additive difference structure.
Then there exists a unique $r \geq 1$ and real-valued functions φ_i, defined on A_i, $i = 1, \ldots, n$, that are unique up to a positive linear transformation, such that, for all $a, b, c, d \in A$,

$$(a, b) \succsim (c, d) \iff \delta(a, b) \geq \delta(c, d),$$

where

$$\delta(a, b) = \left[\sum_{i=1}^{n} |\varphi_i(a) - \varphi_i(b)|^r \right]^{1/r}.$$

Proof. Der Beweis passiert in mehreren Schritten. Zuerst sagen wir etwas über die Definitions- und Zielbereiche der Funktionen F_i und finden damit Grenzen für den Definitionsbereich von F. Danach argumentieren wir, dass wir F bis auf die positiven reellen Zahlen $[0, \infty)$ erweitern können und dass diese Erweiterung eine Funktionsgleichung impliziert, die nur von $f(x) = kx^p$ erfüllt wird. Im letzten Schritt zeigen wir, dass F auf ganz $\mathbb{R}^+$ gleich der Erweiterung sein muss und damit δ eine l^p Metrik sein muss.

Todo: Übersicht vom Beweis aufschreiben

$$F^{-1} \left[\sum_{i=1}^{n} F(t\alpha_i) \right] = tF^{-1} \left[\sum_{i=1}^{n} F(\alpha_i) \right] \tag{6}$$

Todo: necessary sufficient stuff benutzen und zeigen, dass wir das Lemma nutzen können!

Part 1.
By theorem 5.1, the additive difference model

$$\delta(a, b) = G \left(\sum_{i=1}^{n} F_i \left(|\varphi_i(a_i) - \varphi_i(b_i)| \right) \right)$$

yields a metric (that is additive on the coordinate axes) if $F_i(\alpha) = F(t_i\alpha)$ and $F = G^{-1}$ for $i = 1, ..., n$. Thus, F^{-1} is defined for all numbers $\sum_{i=1}^{n} F_i \left(|\varphi_i(a_i) - \varphi_i(b_i)| \right)$, and each F_i coincides with F on its domain (except for the factor t_i, which scales the argument and adjusts the domain).
Since the domains of F_i are intervals (because δ is a metric with additive segments) Todo: begründung checken , we can choose $\omega > 0$ such that for each $i = 1, \ldots, n$ and any α with $0 \leq \alpha \leq \omega/t_i$, α lies within the domain of F_i.
In particular, if we set $\omega_i = \omega/t_i$, then $F^{-1} \left[\sum_{i=1}^{n} F_i(\omega_i) \right] = F^{-1} \left[nF(\omega) \right] =: \Omega$ is defined. Thus, $F = G^{-1}$ is defined at least on the interval $[0, \Omega]$. We can now extend F as follows.

Part 2.
For $\alpha < \omega$, we have by eq. (6),

$$
\begin{aligned}
F^{-1}[nF(\alpha)] &= F^{-1}[nF(\alpha\omega/\omega)] \\
&= (\alpha/\omega)F^{-1}[nF(\omega)] \\
&= \alpha\Omega/\omega.
\end{aligned}
$$

Thus F satisfies the following, by plugging in the correct values for α ($\tilde{\alpha} = \alpha\frac{\omega}{\Omega}$ and $\tilde{\alpha} = F^{-1}(\frac{\alpha}{n})$)

$$
F(a) = nF(a\omega/\Omega), \quad 0 \le a \le \Omega; \tag{7}
$$

$$
F^{-1}(a) = (\Omega/\omega)F^{-1}(a/n), \quad 0 \le a \le nF(\omega). \tag{8}
$$

These two equations can now be used to extend the domain of F and F^{-1}.
We use eq. (7) by defining F through the right-hand side (for values where the right-hand side is defined, i.e., for $0 \le a \le \Omega^2/\omega$). Since $\Omega^2/\omega > \Omega$, this indeed extends F beyond the interval $[0, \Omega]$.
Similarly, F^{-1} satisfies equation eq. (8) for the extended interval $0 \le a \le nF(\Omega) = n^2F(\omega)$.
Moreover, eq. (6) is now valid for $0 \le t\alpha_i \le \Omega$:

$$
\begin{aligned}
F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} nF(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (7))} \\
&= F^{-1}\left[n\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] \\
&= \frac{\Omega}{\omega}F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (8))} \\
&= \frac{\Omega}{\omega}tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] \\
& && \text{(by homogeneity applied for } \alpha_i\omega/\Omega \le \omega) \\
&= tF^{-1}\left[n\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] && \text{(by eq. (8))} \\
&= tF^{-1}\left[\sum_{i=1}^{n} nF(\alpha_i\omega/\Omega)\right] \\
&= tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] && \text{(by eq. (7))}
\end{aligned}
$$

Repeatedly applying the arguments, extends the intervals in which F is defined and in which homogeneity (eq. (6)) is valid by a factor of Ω/ω each time. Thus, we can extend

F to $[0, \infty)$ while preserving homogeneity for $0 \leq t \leq 1$. Further, the derived equation must also hold for $t \in [0, \infty)$. Let $a_i > 0$ and $t > 1$. Then there exists $t' = \frac{1}{t} < 1$ such that $t' \cdot ta_i = a_i$. Therefore

$$\begin{aligned} tt'F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] && (tt' = 1) \\ &= F^{-1}\left[\sum_{i=1}^{n} F(tt'\alpha_i)\right] && (tt' = 1) \\ &= t'F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] \end{aligned}$$

The last step holds, because both $t' < 1$ and $ta_i < \infty$ hold. Dividing both sides by t' yields

$$F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] \qquad \text{for } a, t < \infty \tag{9}$$

The function obtained through this extension must therefore coincide with the original F on at least the interval $[0, \Omega)$.

Part 3.
Next, we derive a functional equation by applying eq. (8) on eq. (9).
Firstly, applying eq. (8) on the left hand side yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right].$$

Applying eq. (8) once more on the right hand side, yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] = \Omega/\omega tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right].$$

Hence

$$F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right].$$

must hold.

According to Theorem 84 (Hardy et al., 1952), the only monotonic functions satisfying this equation for $0 \leq \alpha < \infty$ are functions of the form $k\alpha^r$ (see also Aczél, 1966, p. 153).

Therefore, $F(\alpha) = k\alpha^r$ must hold on at least the interval $[0, \Omega)$.

Part 4.
Now, for any $|\varphi_i(a_i) - \varphi_i(b_i)|$, we can find $t > 0$ small enough such that $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega, i = 1, \ldots, n$. Applying monotonicity yields (we can assume $t \leq 1$):

$$
\begin{aligned}
\delta(a,b) &= F^{-1}\left(\sum_{i=1}^{n} F\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}F^{-1}\left(\sum_{i=1}^{n} F\left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}\left(\sum_{i=1}^{n} \left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)^r\right)^{1/r} && (\text{ since } t|\varphi_i(\alpha_i) - \varphi_i(\beta_i)| \leq \omega) \\
&= \left(\sum_{i=1}^{n} |\varphi_i(a_i) - \varphi_i(b_i)|^r\right)^{1/r}
\end{aligned}
$$

Thus, δ is an l^p metric and, by Theorem Todo: ref add-diff-prox-struct-repr , $F = k \cdot \alpha^r$ and hence $F_i = k(t_i\alpha)^r$ or $F_i(\alpha) = k\alpha^r$ if we incorporate the t_i into the φ_i. Finally, the restriction $1 \leq r < \infty$ follows from the triangle inequality or, more precisely, from Minkowski's inequality.

□

6 Discussion

References

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